

PELKOSENNIEMI PYHATUNTURI, Finland

WMO: 027050

Lat: 67.022N

Lon: 27.218E

Elev: 1599

StdP: 13.87

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-3.3	-11.2	3.2	-6.1	-7.6	3.9	-2.4	53.2	17.6	49.6	18.3	19.5	180	1.086

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	74.1	59.5	70.0	57.7	66.7	56.0	62.2	69.7	59.9	67.8	57.7	64.2	14.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	80.0	65.0	56.8	73.0	61.9	54.9	68.0	59.8	28.8	70.8	27.0	68.1	25.6	64.3	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
46.2	41.8	37.8	DB	-11.5	76.1	3.6	5.0	-14.1	79.7	-16.2	82.6	-18.2	85.4	-20.8	89.0
			WB	-10.2	64.6	5.2	2.9	-13.9	66.6	-17.0	68.3	-19.9	69.9	-23.7	72.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	12.4	14.3	19.6	29.2	41.0	49.5	56.7	52.8	44.4	30.7	23.1	18.5
	DBStd	16.79	8.49	9.25	6.86	6.59	9.36	8.28	7.77	6.56	6.23	6.45	6.61	7.58
	HDD50	6780	1165	999	942	623	311	107	25	43	185	598	806	976
	HDD65	11782	1630	1419	1407	1073	744	469	277	384	619	1063	1256	1441
	CDD50	501	0	0	0	0	33	92	232	128	16	0	0	0
	CDD65	29	0	0	0	0	1	5	19	4	0	0	0	0
	CDH74	108	0	0	0	0	3	15	73	17	0	0	0	0
	CDH80	5	0	0	0	0	0	0	5	0	0	0	0	0
Wind	WSAvg	19.7	22.2	23.1	23.2	18.6	17.9	16.8	15.0	16.8	18.8	19.3	21.9	23.4
Precipitation	PrecAvg	22.2	1.5	1.3	1.2	1.3	1.7	2.4	3.0	2.4	2.1	2.0	1.8	1.6
	PrecMax	29.0	2.8	2.9	2.6	3.1	3.4	4.0	4.6	4.5	4.3	4.0	2.8	2.8
	PrecMin	16.9	0.6	0.5	0.2	0.4	0.5	0.6	0.8	0.7	0.3	0.6	0.6	0.4
	PrecStd	3.1	0.6	0.6	0.6	0.6	0.7	0.9	1.0	1.0	1.0	0.8	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	34.8	36.2	53.6	72.0	76.2	80.0	76.6	60.4	46.4	39.5	33.0
		MCWB	31.3	33.2	32.9	43.4	53.6	60.0	62.9	63.7	53.1	45.5	38.7	32.4
	2%	DB	29.0	30.9	33.0	45.8	66.6	70.5	75.6	69.7	57.4	43.3	36.2	31.1
		MCWB	28.4	30.3	31.0	39.2	52.1	57.7	60.2	58.5	51.5	41.7	35.5	30.5
	5%	DB	26.7	29.2	30.6	40.9	60.6	65.8	71.8	66.0	55.0	41.0	33.5	29.8
		MCWB	26.3	28.9	29.4	35.4	49.3	54.4	58.6	56.5	50.4	39.7	33.3	29.3
	10%	DB	24.1	26.8	28.9	37.8	55.0	61.7	68.2	62.0	52.5	39.2	31.6	28.4
		MCWB	23.7	26.4	28.0	33.4	46.3	52.0	57.3	54.2	49.1	38.4	31.3	28.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.0	33.8	33.6	44.2	54.9	62.7	65.6	64.8	56.7	46.3	38.9	32.6
		MCDB	32.3	35.0	35.3	52.2	66.8	71.4	72.7	74.2	58.7	46.3	39.8	32.8
	2%	WB	28.5	30.5	31.3	39.7	53.0	59.6	62.9	60.8	53.1	42.0	35.4	30.6
		MCDB	29.1	30.7	32.4	45.5	64.6	69.5	69.5	67.1	55.4	42.7	35.9	30.9
	5%	WB	26.3	29.0	29.7	36.1	50.6	56.5	60.9	57.8	51.5	40.0	32.7	29.4
		MCDB	26.7	29.3	30.5	40.1	60.1	62.7	68.7	62.7	53.9	40.9	33.0	29.7
	10%	WB	23.7	26.6	28.2	34.0	47.1	53.4	59.1	55.9	49.8	38.4	31.1	27.9
		MCDB	24.0	26.8	28.8	37.1	54.5	59.2	67.0	60.4	51.6	39.0	31.2	28.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	7.2	7.8	8.8	11.8	11.8	11.7	10.0	7.6	5.6	6.1	6.4
		MCDBR	10.1	8.4	8.3	11.7	17.9	16.0	14.9	13.6	10.3	7.4	7.4	7.0
		MCWBR	10.0	8.2	7.3	7.0	9.6	7.9	6.8	7.0	6.8	6.4	7.8	7.1
	5% WB	MCDBR	10.1	8.2	7.3	10.8	17.1	14.8	13.7	11.4	9.0	7.2	7.4	7.1
		MCWBR	10.1	8.0	6.8	7.1	10.0	8.5	7.2	7.1	6.6	6.5	7.4	7.1
Clear-Sky Solar Irradiance	taub	0.180	0.245	0.278	0.309	0.324	0.326	0.338	0.321	0.298	0.267	0.348	N/A	
	taud	1.923	2.118	2.198	2.254	2.471	2.505	2.483	2.508	2.555	2.355	2.777	N/A	
	Ebn at Noon	61	188	246	269	279	280	272	266	246	185	71	0	
	Edh at Noon	4	16	25	32	29	29	28	25	19	13	5	0	
All-Sky Solar Radiation	RadAvg	19	177	651	1228	1520	1619	1492	1089	566	190	37	0	
	RadStd	1	28	59	99	133	125	150	102	56	36	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)		+1.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page